Lesson 4: SECURITY SERVICES

Duration: 15 hours

Learning Outcomes

At the end of this lesson, students should be able to:

1. Describe possible availability levels for a web service.
2. Explain how redundancy and geographic dispersion relate to availability.
3. Explain the roles of integrity, confidentiality, availability, authentication, and non-repudiation as security services.
4. Explain how one-way cryptographic functions are used to implement integrity in document transfer.
5. Explain how cryptographic encryption algorithms are used to implement confidentiality in document transfer.
6. Explain how one-way functions and encryption are used to implement authentication services.
7. Explain how cryptographic functions are used to implement non-repudiation services.

Module Overview

Security services are fundamental mechanisms that protect information systems against threats and attacks. These services ensure that information remains accessible to authorized users, accurate and trustworthy, protected from unauthorized disclosure, and that users can be reliably identified and held accountable for their actions. This module provides an in-depth discussion of the core security services and explains how cryptographic techniques support secure communication and system operations.

Topic 1: Overview of Security Services

Discussion

A **security service** is a set of techniques and controls designed to protect information and information systems. These services address risks across all information states—storage, transmission, and processing.

Core Security Services

1. Availability
2. Integrity
3. Confidentiality
4. Authentication (Source Reliability)
5. Non-repudiation

Topic 2: Availability

Discussion

Availability ensures that systems, services, and data are accessible to authorized users when needed.

Availability Levels for Web Services

- **Best Effort Availability** – No formal uptime guarantee
- **High Availability (HA)** – Typically 99.9% to 99.99% uptime
- **Fault-Tolerant Availability** – Near-continuous operation, even during failures
- **Mission-Critical Availability** – 99.999% (“five nines”) or higher

Example

An e-commerce website aims for high availability to avoid revenue loss during outages.

Topic 3: Redundancy and Geographic Dispersion

Discussion

Redundancy involves duplicating critical components to prevent single points of failure.

Geographic dispersion places system components in different physical locations.

Examples

- Multiple servers behind a load balancer
- Data centers in different regions or countries

Benefits

- Improved fault tolerance

- Protection from natural disasters
- Reduced downtime

Topic 4: Integrity

Discussion

Integrity ensures that information is accurate, complete, and unaltered during storage, processing, or transmission.

Cryptographic Integrity Mechanisms

- One-way hash functions (SHA-256, SHA-3)
- Message Authentication Codes (MAC)
- Digital signatures

Example

A downloaded file is verified by comparing its hash value with the original.

Topic 5: Confidentiality

Discussion

Confidentiality ensures that information is accessible only to authorized individuals.

Cryptographic Techniques

- Symmetric encryption (AES)
- Asymmetric encryption (RSA, ECC)
- Hybrid encryption systems

Example

HTTPS encrypts data transmitted between a browser and a web server.

Topic 6: Authentication (Source Reliability)

Discussion

Authentication verifies the identity of a user or system.

Authentication Mechanisms

- Passwords and PINs
- Cryptographic hashes for password storage
- Digital certificates and public key infrastructure (PKI)

Example

A user logs in to an online banking system using a password and one-time code.

Topic 7: Non-Repudiation

Discussion

Non-repudiation prevents parties from denying actions they have performed.

Cryptographic Tools

- Digital signatures
- Public key infrastructure (PKI)
- Secure audit logs

Example

A digitally signed contract ensures the sender cannot deny signing it.

Topic 8: Security Services in Practice

Integrated Example

When sending a secure email:

- Confidentiality: Message is encrypted
- Integrity: Hash ensures content is unchanged

- Authentication: Digital certificate verifies sender
- Non-repudiation: Digital signature proves authorship
- Availability: Email servers ensure delivery
